

Maser Catalog Analysis with Astropy

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Introduction

This document demonstrates how to create a maser catalog with real 6.7 GHz methanol maser sources, load it using **Astropy**, extract coordinates, and visualize them. The methodology follows the workflow of a Jupyter notebook.

1. Creating the Maser Catalog

The maser catalog includes real sources from *Ellingsen et al., 1996*. The table below lists the sources and their coordinates:

Source	RA (deg)	Dec (deg)	Reference
G9.62-0.02	9.6233	-0.0204	Ellingsen et al., 1996
G10.47-0.03	10.4742	-0.0332	Ellingsen et al., 1996
G11.11-0.12	11.1105	-0.1248	Ellingsen et al., 1996
G12.80-0.20	12.8042	-0.2056	Ellingsen et al., 1996
G13.88-0.29	13.8800	-0.2900	Ellingsen et al., 1996
G15.05-0.68	15.0500	-0.6800	Ellingsen et al., 1996
G16.59-0.05	16.5900	-0.0500	Ellingsen et al., 1996
G18.67-0.03	18.6700	-0.0300	Ellingsen et al., 1996
G19.61-0.23	19.6100	-0.2300	Ellingsen et al., 1996
G20.60-0.23	20.6000	-0.2300	Ellingsen et al., 1996

The catalog can also be saved as a CSV file and loaded for analysis.

2. Loading the Catalog with Astropy

The following Python code demonstrates loading the catalog using Astropy's **Table** module:

```
from astropy.table import Table

# Load the catalog
catalog = Table.read('real_maser_catalog.csv', format='csv')

# Display the first few rows
print(catalog[:5])
```

3. Extracting Coordinates

To extract RA and Dec coordinates and convert them into SkyCoord objects:

```
from astropy import units as u
from astropy.coordinates import SkyCoord

ra_values = catalog['RA']
dec_values = catalog['Dec']

coords = SkyCoord(ra=ra_values*u.deg, dec=dec_values*u.deg)

for i, c in enumerate(coords):
    print(f"{i+1:02d}: RA = {c.ra.to_string(unit=u.hour, sep=':')}
          ", "
          f"Dec = {c.dec.to_string(unit=u.deg, sep=':')}")
```

4. Visualizing the Maser Coordinates

The RA and Dec coordinates can be visualized using a scatter plot:

```
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))
plt.scatter(coords.ra.deg, coords.dec.deg, color='blue', s=50)
for i, src in enumerate(catalog['Source']):
    plt.text(coords.ra.deg[i]+0.1, coords.dec.deg[i]+0.01, src,
             fontsize=8)
plt.xlabel('Right Ascension (deg)')
plt.ylabel('Declination (deg)')
plt.title('Maser Source Coordinates')
plt.grid(True)
plt.show()
```

Conclusion

This workflow demonstrates how to:

- Create a real maser catalog with RA/Dec coordinates.
- Load and manipulate the catalog using Astropy.
- Extract and work with astronomical coordinates using SkyCoord.
- Visualize the maser sources in RA-Dec space.

This LaTeX document mirrors the structure and functionality of the Jupyter notebook, making it suitable for lecture notes or a report.